

Uncertainty Quantification for Predicting Stellar Properties from Gaia Spectra

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Abstract

We compare a linear regression baseline to three different neural network-based architectures: ensemble networks for pure regression, dropout-induced networks producing Gaussian standard deviation and mean estimators, and simulation-based inferencing (SBI) via neural posterior estimation. Uncertainty quantification and model adjustments with fudge factors are then performed manually alongside automatic adjustments with conformal prediction. The three neural network methods all surpass the linear regression baseline in all prediction metrics (RMSE, R^2) and have lower uncertainty (tighter median interval widths and better conditional coverage). We conclude that neural network-based estimators far surpass a linear regression baseline and provide tighter uncertainty quantification.

Objectives

In order to to produce an honest account of uncertainty and limitations in predicting stellar metallicity properties, we wanted to

- Build ML models that predict a star's physical properties (temperature, gravity, composition) from its Gaia spectrum
- Attach a calibrated uncertainty to every prediction

Our framework:

- Explain how a linear regression regression baseline model performs for predicting star metallicity $[Fe/H]$
- Produce 3 higher-complexity architectures involving neural nets to implement the same predictions
- Assess each model to find whether aleatoric or epistemic uncertainty is present in our processes
- Increase conditional model coverage on average across all star groups, with manual methods involving finding a fudge factor

Methods

Umang:

- Trained a model using ensembling
- Models mostly accounted for aleatoric uncertainty
- Assuming $\varepsilon \sim \mathcal{N}(\mu, \sigma^2)$, intervals were calculated using an estimated standard deviation and a fudge factor was tuned to attain a CI
- Score function used in conformal prediction was MAE

Johan:

- Trained a neural net that also predicts standard deviation of errors $sd(\varepsilon)$, assuming Gaussian errors
- Used Gaussian negative log-likelihood loss
- Use the z -score as the score function in conformal prediction, with intervals from $\alpha/2$ to $1 - \alpha/2$ quantiles

Edward:

- Trained a neural net to predict posterior probabilities using SBI with NPE normality flow, resembling shifted and skewed Gaussians
- SBI NPE maximizes negative log likelihood loss
- Assumed a uniform prior for the simulations
- Used mean from 200 samples on posterior distribution as predicted value

Umang's Findings

- Averaging five seeded neural networks improves point prediction accuracy, reducing RMSE from 0.0985 to 0.0953 dex.
- Compared with the baseline RMSE of roughly 0.2 dex, the neural network ensemble nearly halves the prediction error.
- The ensemble maintains comparable coverage while reducing median interval width by approximately 35%, yielding more informative uncertainty estimates.
- Uncertainty increases toward low metallicity, suggesting that metal-poor stars are harder to predict reliably.
- Conformal prediction achieves approximately 90% marginal coverage, though coverage is lower for metal-poor stars and giants than for metal-rich dwarfs.

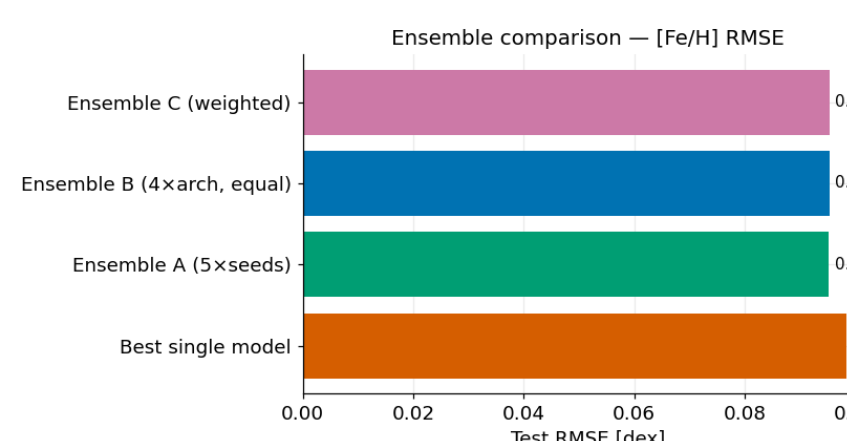


Figure 1. Test RMSE Comparison of Neural Network Ensembles

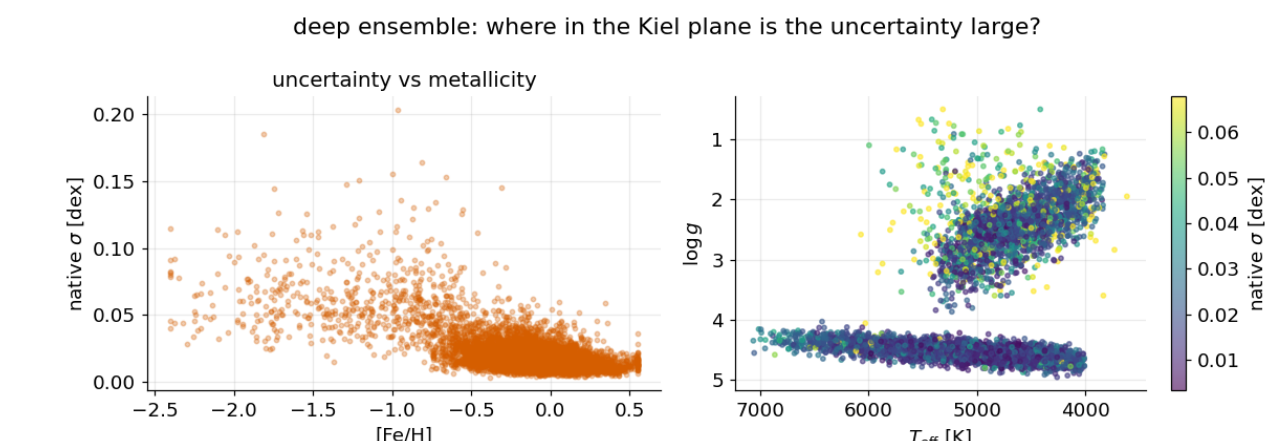


Figure 2. Where Does the Neural Network Become Uncertain?

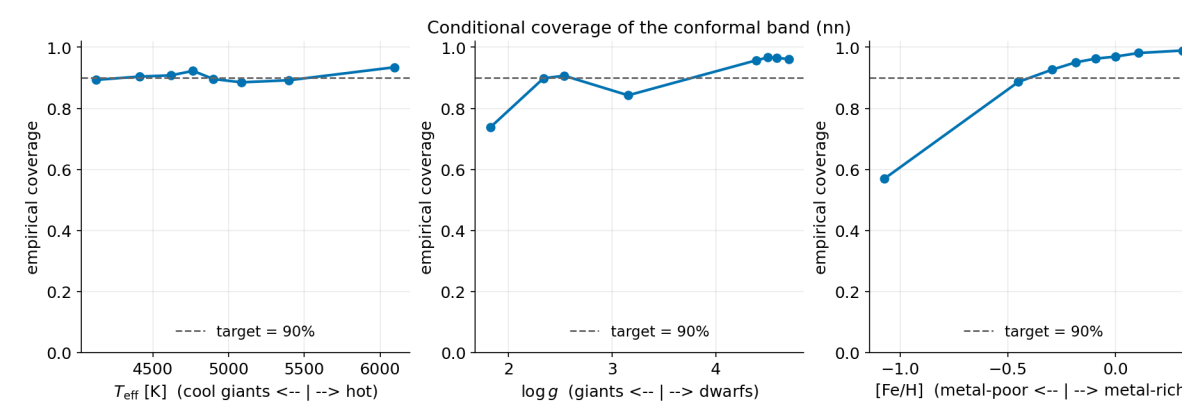


Figure 3. Coverage Varies Across Stellar Parameter Space

Johan's Findings

- The Gaussian neural network produces adaptive uncertainty estimates that achieve good conditional coverage on the calibration set.
- Unlike the shared-floor baseline, which assigns the same uncertainty to every star, the predicted standard deviation better adapts to easy and difficult cases.
- This substantially reduces median interval width relative to the constant-width baseline, from roughly 0.286 dex to 0.168 dex.
- Conformal calibration slightly improves coverage, from 90.27% to 90.68%, with only a small increase in median interval width from 0.168 to 0.171 dex.

Method	Coverage	Median Width
Native	0.9027	0.1681
Conformal	0.9068	0.1706

Table 1. Coverage and median interval width for native Gaussian neural network uncertainty and conformalized intervals.

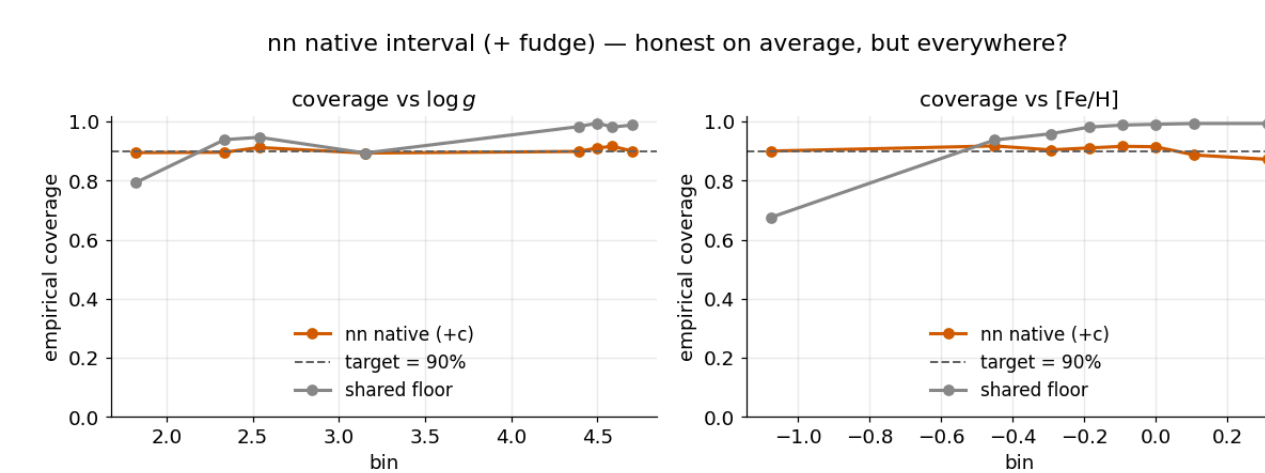


Figure 4. Conditional Coverage of Adaptive vs. Constant

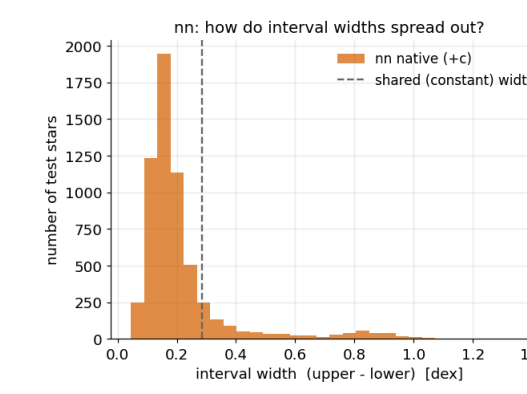


Figure 5. Uncertainty Interval Widths

Edward's Findings

- Predictive uncertainty is summarized by $\sigma = (q_{84} - q_{16})/2$ rather than the standard deviation, making it more robust to non-Gaussian tails.
- The mean uncertainty is 0.076 dex, with larger values for metal-poor stars and underrepresented regions of the training data.
- A calibration factor of $c = 1.1$ gives near-nominal coverage, with 91.4% calibration coverage and 91.3% test coverage.
- Native NPE intervals are substantially tighter than the shared-floor baseline, with median widths of 0.211 dex versus 0.337 dex.
- Across the Kiel plane, uncertainty reflects data density: dwarfs are better constrained than giants.
- The uncertainty is interpreted as mostly epistemic, although an aleatoric floor of roughly 0.07–0.08 RMSE may remain.

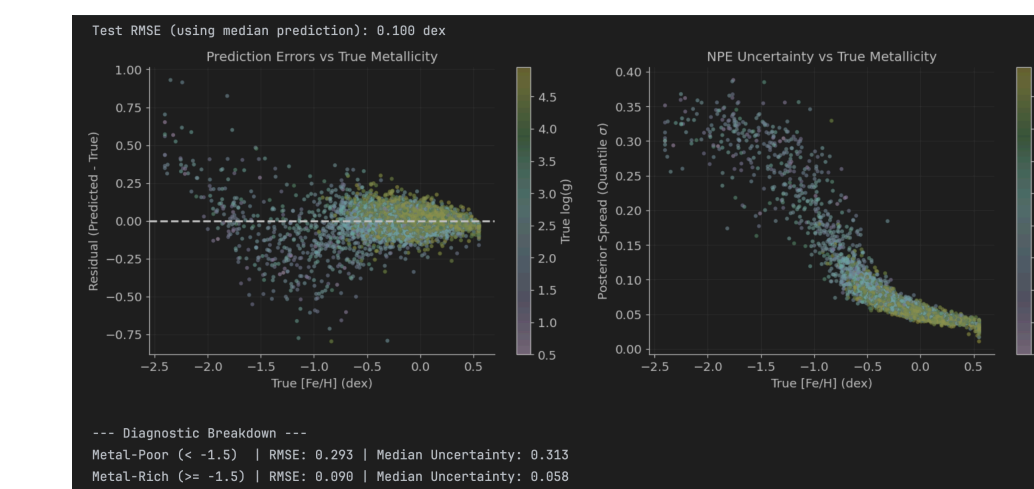


Figure 6. Residuals and Uncertainty

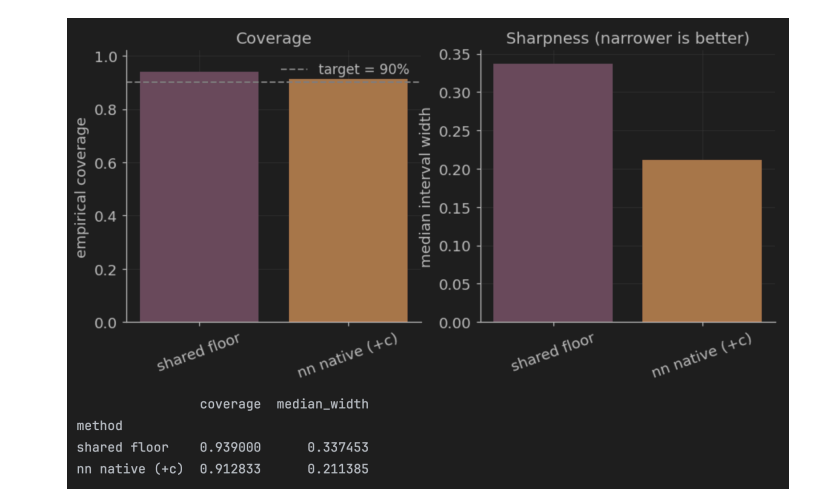


Figure 7. Coverage and Sharpness

Results

Model	RMSE	R^2	FF c	FF Width	Conf. Width
Umang (Ensemble)	0.095	0.948	3.90	0.206 dex	0.259 dex
Johan (Gaussian Prediction)	0.087	0.960	1.00	0.168 dex	0.171 dex
Edward (SBI NPE)	0.103	0.943	1.10	0.211 dex	0.265 dex

Table 2. Model comparison. RMSE and R^2 are on the test set. c is the fudge factor. FF Width and Conf. Width are the fudge factor and conformal median interval widths respectively.

Conclusions

- Neural net-based methods universally performed better than linear regression baseline
- Aleotoric uncertainty was present in all processes, epistemic present in the distribution-based methods (Johan, Edward)

References

- Anastasios N. Angelopoulos and Stephen Bates. Conformal prediction: A gentle introduction. *Found. Trends Mach. Learn.*, 16(4):494–591, March 2023.
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